CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EC) Examination

May 2014

EC405 Digital System Design

Date: 24.05.2014, Saturday   Time:01.30p.m. To 4.30p.m.   Maximum Marks: 70

Instructions:

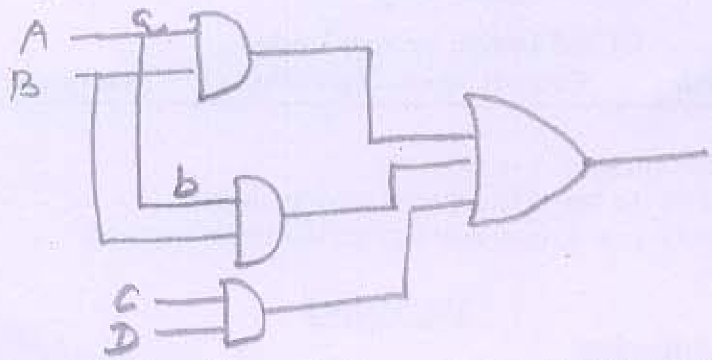
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION-I

Q-1	**Attempt the following.**	
1	For NAND gate all the input s-a-0 faults and the output _______ faults are equivalent.	**01**
2	The Boolean expression (a'+b')(b+c) contains static 0 or static 1 hazard?Justify your answer.	**02**
3	Define the following terms:Aliasing, Fault Dominance	**02**
4	Decade counter can be realized with a 22V10 SPLD or not? Justify your answer.	**02**
Q-2(a)	A sequential network has one input (x) and one output (z). The output z=1 if the total number of 1's received is divisible by 3(Note:0,3,6,9,... are divisible by 3)	**04**
(b)	Draw a block diagram of a 4-bit parallel multiplier. Explain operation of it and also describe control state graph.	**04**
(c)	Find a minimum-row PLA table to implement the following sets of Functions. F1(a,b,c,d)= Σm(3,4,6,9,11) F2(a,b,c,d)= Σm(2,4,8,10,11,12) F3(a,b,c,d)= Σm(3,6,7,10,11)	**06**
	OR	
Q-2 (a)	A sequential network has one input (x) and one output (z). The output z=1 if the total number of 1's received is odd and at least two consecutive 0's have been received. Find Moore state graph.	**04**
(b)	What is Fault Dominance? Find out fault equivalence class for below mentioned gates. NAND,NOR	**04**
(c)	Using the method of map entered variables, use 4 variable maps to find a minimum sum of product expression. Also verify it with 5 variable kmap F(A,B,C,D,E)=∑m(0,4,5,7,9)+ ∑d(6,11)+E(m1,m15)	**06**

Q-3(a) Which are the main modules of keypad scanner? Explain the working of key scan using state graph. **06**

(b) Find a minimum set of test vectors that will detect the fault a s-a-0 in the following network. **06**

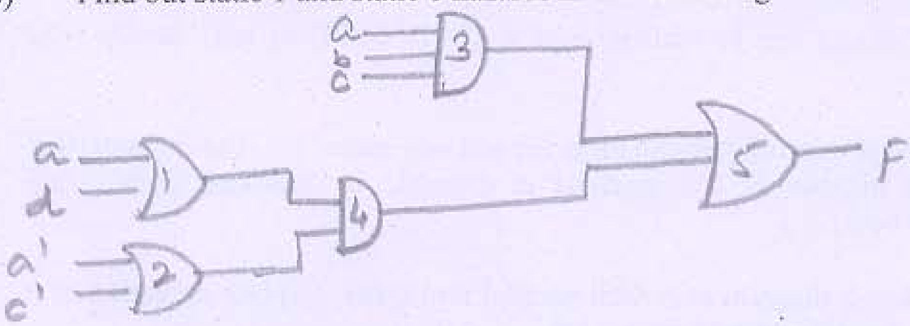


(c) Which are the different types of pattern generators for BIST architecture? **02**

OR

Q-3 (a) Draw a block diagram of a 4 bit serial adder with Accumulator and Design control circuit for it. **06**

(b) Find out static 1 and static 0 hazards in the network given below if any. **06**



(c) Explain the steps for sequential design process. **02**

SECTION II

Q-4 **Do as directed.**

1 What is the difference between melay and moore sequential circuit? **02**

2 Define the following terms: **02**
Bridging faults, Signature analysis

3 For NOR gate input s-a-1 then remove ________ **01**

4 What is the main difference between a PAL- and a GAL-based SPLD? **02**

Q-5(a) Explain antifuse and SRAM programmable switching technologies in detail. **06**

(b) Given the polynomial $P(x) = x^8+x^7+x^2+1$ **05**
Design a LSFR with characteristic polynomial $P(x)$.

	(c)	Explain the component of SM chart.	03
		OR	
Q-5	(a)	Design binary divider to divide an 8 bit number by a 4 bit number to find a 4 bit quotient. The right 4 bit of the dividend register should be used to store the quotient. Draw a block diagram for the divider.	06
	(b)	Draw and explain the SM chart for UART transmitter.	05
	(c)	Design Melay state machine for 1010.	03
Q-6	(a)	Compare the different programming technologies used in PLD.	04
	(b)	What is Built-In-Self-Test? Discuss the issues and benefits of BIST. Describe BIST architecture and its operation.	07
	(c)	Alyssa P. Hacker has a snail that crawls down a paper tape with 1's and 0's on it. The snail smiles whenever the last five digits it has crawled over are 101. Design Moore FSM of the snail's brain.	03
		OR	
	(a)	What is BILBO? Describe BILBO architecture and its operation?	06
	(b)	Explain the SM chart for Dice game and implement it using PLA and flip-flop.	08